# AI Dub & Filter Studio — Product Concept

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## Overview

A web app where people upload a movie or TV episode and get back a personalized version: dubbed by AI into any language, with subtitles in any language and style, and with objectionable scenes (nudity, profanity, violence) filtered out to their own standards.

Today these three needs are served separately and poorly. Official dubs exist for only a handful of languages per title, fan subtitles are inconsistent, and family-filtering tools cover a limited catalog. One pipeline that understands a video's speech, faces and scenes can deliver all three from a single upload.

## Target users and use cases

| User | What they want | Key features |
|---|---|---|
| Families with kids | Watch mainstream films without nudity or crude language | Filtering, profiles per child |
| Multilingual households | Grandparents and kids watching the same show in different languages | Dubbing, dual subtitles |
| Language learners | Watch in a target language with native-language subtitles | Dubbing, bilingual subtitles |
| Religious and conservative viewers | Consistent content standards across everything they watch | Filter presets, shared standards |
| Schools and libraries | Show films in class without inappropriate scenes | Filtering, audit reports |

The product is for viewers: people who want to clean up and dub the movies and shows they watch. Families are the core launch audience, since filtering is the clearest reason to pay.

## Feature 1: AI dubbing in any language

The user picks a target language and gets a new audio track where each character speaks that language in a voice that matches the original actor's tone, timing and emotion.

How it works for the user:

- **Pick languages** — one or several target languages per title; switch between them during playback like any audio track.
- **Voice style** — "match original voices" (voice cloning per character) or "neutral narrator voices" (stock voices, cheaper and fewer rights issues).
- **Speaker labels** — the system detects each character; users can rename them ("Speaker 2" → "Walter") and assign a voice.
- **Translation tone** — literal, natural, or kid-friendly (which also softens language, tying into filtering).
- **Lip-sync (premium)** — optionally re-render mouth movements so on-screen lips match the new language.
- **Script editor** — view and edit the translated script line by line, then re-generate only that line's audio.
- **Background preservation** — music and sound effects are separated from dialogue and kept, so only the voices change.

Quality bar: dubbed lines should fit the original line's duration within a small tolerance, so translation is length-aware (it rephrases to fit rather than speeding up speech).

## Feature 2: Custom subtitles

Subtitles are generated from the same transcript that drives dubbing, so they are available in any language at no extra processing cost.

- **Any language, any combination** — subtitle language is independent of audio language (e.g. Spanish audio with English subtitles).
- **Dual subtitles** — two languages stacked on screen, aimed at language learners.
- **Subtitle types** — dialogue only, full closed captions (sound cues like [door slams]), or forced-narrative only (signs, foreign-language lines).
- **Styling** — font, size, color, background box, position, and reading-speed limits for kids or accessibility.
- **Filtered subtitles** — bleeped words appear as "****" or a clean substitute, matching the filtered audio.

- **Editing and upload** — users can fix lines in an editor, or upload their own SRT/VTT file instead.
- **Export** — download as SRT or VTT for use in other players.

## Feature 3: Content filtering

AI scans every frame and line of dialogue, tags objectionable moments with a category and severity, and applies the user's chosen action during playback. Each user sets their own standard rather than relying on a single rating.

### What gets detected

| Category | Examples | Severity levels | Default action |
|---|---|---|---|
| Profanity | Swearing, slurs, crude humor | Mild, moderate, strong | Mute or replace word |
| Blasphemy | Religious names used as exclamations | On / off | Mute |
| Nudity | Partial to full nudity | Partial, full | Skip scene or blur |
| Sexual content | Kissing to explicit scenes, innuendo | Romance, suggestive, explicit | Skip scene |
| Violence and gore | Fights, blood, injury detail | Mild, moderate, graphic | Skip or blur |
| Substance use | Drugs, drinking, smoking | On / off | Skip |
| Frightening content | Jump scares, disturbing imagery | Mild, intense | Skip or dim audio |

### Filter actions

- **Mute** — silence just the word (millisecond precision from the transcript).
- **Replace** — swap the word with a clean alternative in the AI voice ("shoot" for a swear), seamless because the same voice model produced the dub.
- **Blur / crop** — blur a region of the frame, or crop to keep only the safe part.
- **Skip** — jump over the scene; a short on-screen note ("Skipped 0:42 — sexual content") is optional.

- **Summarize** — replace a skipped scene with a one-line text card so the plot still makes sense.

**User controls**

- **Presets** — "Kids (G)", "Family (PG)", "Teen (PG-13)", "Language only", "Custom".
- **Per-category sliders** for fine control.
- **Profiles** — different settings per household member, with a PIN-locked parent profile.
- **Preview report** — before watching, a timeline showing what will be filtered and how much runtime is removed.
- **Review and override** — users can flag missed or wrongly filtered moments; those corrections improve future detection.

Filtering never edits the uploaded file. It is a list of timestamps and actions applied by the player, so it can be changed or turned off at any time.

## User flow

1. **Add a title** — upload a video file, or (later) connect a streaming source; confirm the title so metadata and any existing community filter data load.
2. **Choose outputs** — pick dub languages, subtitle languages and style, and a filter preset or profile.
3. **Process** — the video is analyzed once (transcript, speakers, scenes); a progress bar shows each stage. A 2-hour film should be ready in well under its own runtime.
4. **Review** — see the filter timeline and the translated script; adjust anything before watching.
5. **Watch** — the web player streams the video with the chosen audio track, subtitles and filters applied live; viewers can switch language or profile mid-show.
6. **Feedback** — flag a missed scene or a bad translation in one tap; fixes apply immediately and feed model improvement.

Because analysis happens once per title, changing languages or filter settings later is near-instant.

## Content permissions

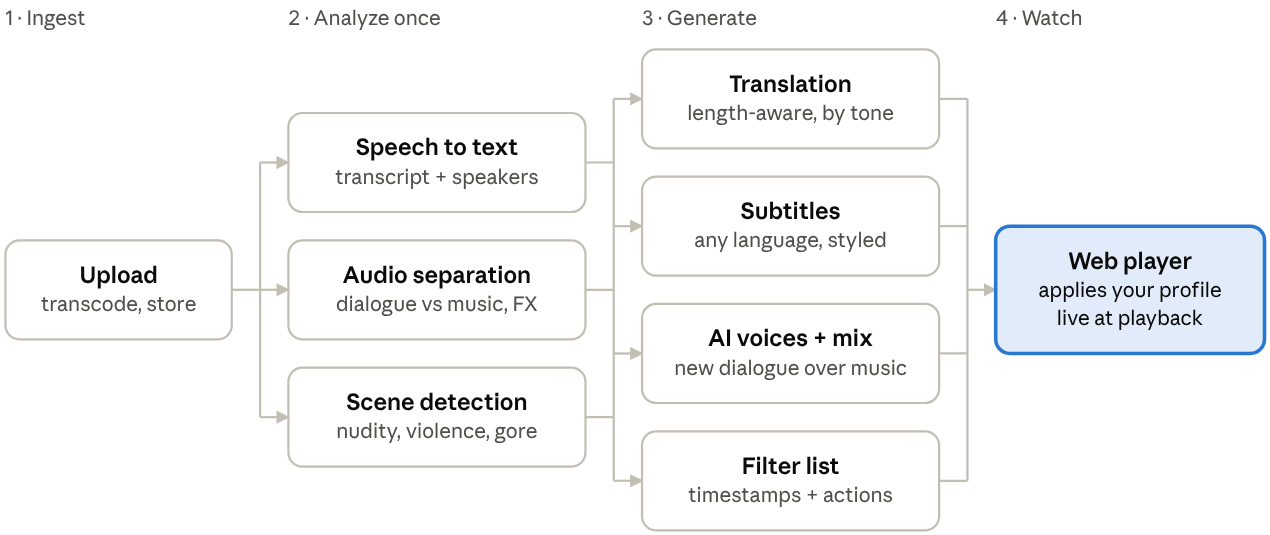
Working assumption: every title on the platform comes with permission for AI dubbing and content filtering. Licensing is handled outside this product scope for now.

- Each title carries a permissions flag (dubbing, filtering) checked before processing.
- Filters stay a playback overlay, so the original file is never altered.

## Technical architecture

Each title is analyzed once; every language, subtitle and filter setting is generated from that shared analysis, and the original file is never altered.

**One analysis pass feeds dubbing, subtitles and filters**




processing pipeline · 4 stages

The player combines the original video, the chosen audio track, subtitles and the filter list at playback time.

| Component | Job | Build or buy to start |
|---|---|---|
| Speech to text + diarization | Word-level transcript with timestamps and speaker labels | Buy (speech API), later self-host an open model |
| Audio source separation | Split dialogue from music and effects | Open-source model on GPU workers |
| Scene detection | Classify frames for nudity, violence, gore; detect scene cuts | Vision moderation API + custom fine-tuning |
| Translation | Context-aware, length-aware script translation | LLM API |

| Component | Job | Build or buy to start |
|---|---|---|
| Text to speech + voice cloning | Emotional, timed voices per character | Voice AI API |
| Lip-sync | Re-render mouths to match new audio (premium) | Buy, later |
| Job pipeline | Queue, GPU workers, retries, progress | Build (queue + cloud GPU) |
| Player | HLS streaming, track switching, live skip/mute/blur | Build on an open-source web player |

## Business model

AI processing (especially voice generation) is the main cost and scales with runtime, so pricing should track minutes processed. Figures below are placeholders to validate.

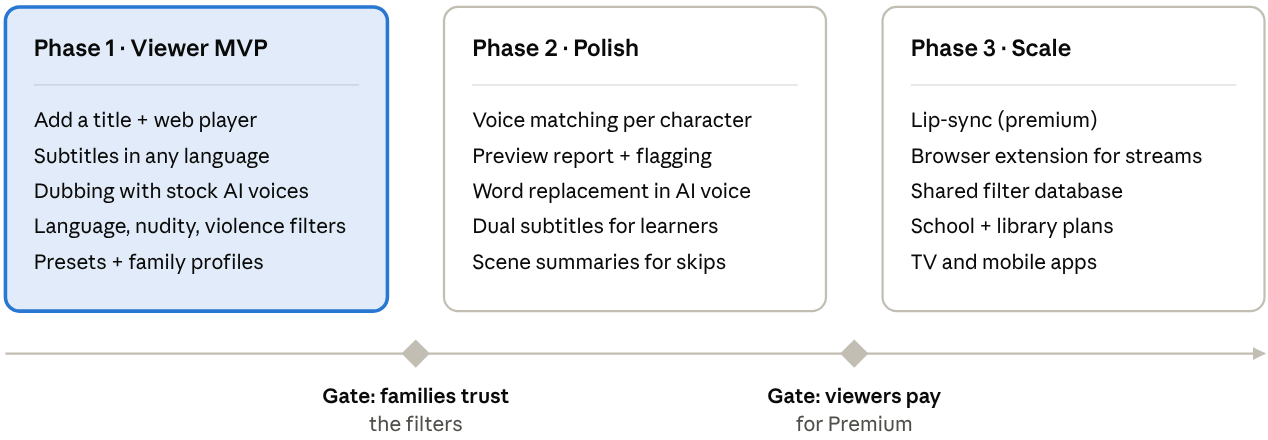
| Plan | For | Includes | Pricing basis |
|---|---|---|---|
| Free | Trying it out | Subtitles + filtering on a few titles, 1 dub language | Capped titles per month |
| Family | Households | Filter profiles, subtitles, monthly dubbing hours | Monthly subscription |
| Premium | Heavy viewers, language learners | Voice matching per character, lip-sync, dual subtitles, unlimited languages | Higher monthly tier |
| Business | Schools, libraries | Bulk access, classroom presets, audit reports | Annual contract |

A second revenue stream is the filter database itself: once a title is tagged, the tags can be reused by every user of that title, making each new user cheaper to serve.

## MVP scope and roadmap

The MVP targets viewers who want to clean up and dub movies and shows: it ships the full loop — filter, dub, subtitle, watch — on permission-cleared titles.

**Launch the full viewer experience first, then polish, then scale**




roadmap · 3 phases, 2 gates

Filtering ships in Phase 1 because it is the main reason families sign up; voice matching and lip-sync wait until the core experience is trusted.

## Risks and open questions

| Risk | Impact | Mitigation |
|---|---|---|
| Missed nudity or profanity | Parents lose trust after one miss | Conservative thresholds, human review for popular titles, easy flagging |
| Over-filtering ruins the story | Viewers churn | Severity levels, scene summaries, preview report |
| Dub quality feels robotic | Low retention | Emotion-aware voices, per-line regeneration |
| Processing cost per film | Thin margins | Shared filter tags per title, cached translations |
| Storage of large video files | High hosting cost | Process each title once, stream to all viewers |
| Permissions not tracked per title | Content processed without rights | Permissions flag required before processing |

### Open questions

- [ ] Web player only, or browser extension / TV apps too?
- [ ] Build AI models in-house or use third-party APIs to start?
- [ ] Should filter tags be community-reviewed, and how are reviewers trusted?
- [ ] Which languages are in the first release?